2013 17th IEEE International Enterprise Distributed Object Computing Conference Workshops

Towards a Methodology for Goal-Oriented Enterprise Management

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Abstract — **This paper discusses the potential benefits of incorporating a goal-oriented perspective to Business Process Management (BPM) from a methodological point of view, with a focus on the BPM Design and Evaluation phases. To achieve this aim, we examine the current activities of the BPM lifecycle and on the basis of that, the lifecycle is increased with new activities related to the modeling of business services and goals. In the BPM design phase, we suggest the integration of business services, goals and KPI modeling to the previously modeled business processes, also describing how the existing methods in goal and KPI modeling can be used to perform such representation. In the Evaluation phase, we discuss how the previously modeled KPIs can be evaluated in terms of operational data. The information produced by the KPI monitoring enables one to verify whether the business goals are being adequately fulfilled by the enterprise architecture. Further, business services can be evaluated with respect to their ability in adequately aggregating value for the final customer.**

Keywords— goal modeling, service modeling, Key Performance Indicators, methodology, Enterprise Architecture, Business Process Management

I. INTRODUCTION

The increasing competitiveness drives organizations to constantly evaluate their position in the market and promote changes in an attempt to improve the quality of the services and products they offer. In this context, the discipline of Business Process Management (BPM) arises as an effective tool whose main aim is to support the design, administration, configuration, enactment and analysis of business processes [1]. Once business processes capture how the organization translates the corporate strategy into operational aspects, the ultimate goal of the BPM discipline is to promote the management of processes with the aim of better achieving enterprise's goals.

Although much evolution has been achieved in several phases of the BPM lifecycle towards a broader view of process management in recent years, a careful examination of the area evidenced that still too little research is devoted to target efforts related to process improvement [2] during the BPM evaluation phase. This is somehow problematic once processes obviously evolve (improve) to better achieve enterprise goals. Furthermore, few approaches in *process performance management* and BPM address the problem of monitoring process to quantify their progress towards achieving strategic goals, rather taking an operational view for process performance [3]. Third, even among those approaches that recognize the benefits of adopting goal modeling for increasing the value of process models [4] [5], little attention has been devoted to the association between goal models and other elements of the enterprise architecture, such as resources, roles and norms [6] with the aim of improving process operations.

This issue of connecting processes to a more strategic perspective is also addressed by the discipline of Enterprise Architecture (EA) [7] [8] that recognizes the interconnection of the several enterprise's perspectives, such as organizational structure, business processes, information systems, and infrastructure [7] and how these enterprise perspectives can be combined to better expose the organizational services to its customers. Among these perspectives, the domain of goals have been also recognized as an important element of enterprise architectures, since it allows architects to systematically express the reasons for adopting one particular enterprise setting (in face of multiple alternatives) [9]. Nevertheless, goal modeling in the discipline of EA is still incipient as reveled in [10], and the proposals do not deal with the whole *goal lifecycle*, in which goals are initially created, operationalized and evaluated accordingly. Furthermore, EA also relies on the association with other external tools from BPM such those used for evaluation of operational data to promote an adequate enterprise management (and indirectly evaluate the goal satisfaction).

To gain a more strategic perspective within the disciplines of BPM and EA, this paper proposes the enhancement of the current BPM lifecycle with new activities and artifacts related to the management of organizational services and enterprise goals. We start with the view that the enterprise exists to deliver (business) services to its external customers. These services are usually associated with the achievement of goals from both sides, namely, the customers and enterprise. From an internal perspective, processes coordinate work and resources to achieve the enterprise's goals. These enterprise's goals must be measured accordingly through the use of Key Performance Indicators (KPIs) to evaluate whether the operational performance is being able to achieve these goals. This overall idea comprehends what we call *Service-oriented EA* (SoEA) and builds on ideas of the disciplines of BPM and EA with the aim of integrating the operational perspective provided by BPM with the service-, goal-orientation of the EA discipline to gain a more strategic organizational perspective. It is important to mention that we describe a methodology, i.e. "a documented approach for performing activities in a coherent, consistent, accountable,

978-0-7695-5085-5/13 $26.00 © 2013 IEEE
DOI 10.1109/EDOCW.2013.17

and repeatable manner" [11], which guides the usage of service-, goal-oriented techniques that can be used to enhance the BPM lifecycle. The aim is not to give a complete discussion containing all aspects that may ever arise in relation this methodology, but to provide methodological guidance with respect to a number of the new opportunities in the context of goal-oriented enterprise management.

The rest of this paper is organized as follows: Section II initially describes the general overview of our methodology and how it relates to the existent BPM lifecycle. Section III concludes the paper with a general discussion about the methodology enhancement and the future research opportunities regarding the adoption of these new activities within the existent lifecycle.

II. General Overview of a Goal-Oriented Methodology for Business Process Management

Although the nomenclature and number of phases within the BPM lifecycle may vary depending on the author, there is some agreement in relation to the definition of each phase. We ground our considerations based on a lifecycle that is commonly accepted within the BPM community [12] [1].

Basically describing this lifecycle, it starts with the **Business Process Design and Analysis Phase** in which process are identified and modeled. The **Business Process Configuration Phase** addresses the selection of computer systems and their configuration to support the processes that are indeed executed and monitored in the **Business Process Enactment Phase**. Finally, the **Business Process Evaluation Phase** is concerned about using the available information to evaluate and improve process models and their implementations.

As mentioned earlier, this lifecycle focuses on the operational aspect of processes, putting little effort into capturing their strategic perspective. To address that, we describe the BPM lifecycle enhanced with the concepts of business services and goals, that we believe to provide a more strategic view for process, linking their operation to the services exposed by the organization.

In Table I, the introduced activities within the current BPM lifecycle are depicted in conjunction with the already existent ones (the new activities are depicted in boldface). The **Design Phase** of our methodology assumes a *top-down* view that organizations exist to deliver services (to satisfy stakeholders' goals). As a consequence of that, we have to represent these (business) services and goals. Next to that, goals have to be operationalized, i.e., linked to the processes and other *organizational elements* (or *architectural elements*) responsible for their achievement, such as roles, information systems, applications, etc. After a suitable operationalization, goals need to be adequately measured via KPIs. The **Evaluation Phase** on the contrary, assumes a *bottom-up* fashion once it aggregates properties of processes to deliver information about their behavior (in conjunction with the several *organizational elements*).

Table I Phases and steps of our methodology

Methdology	*Steps*
Design Phase (Top-down Methodology)	1. **Model organizational services as stakeholders' goals** 2. **Operationalize goals** 3. Model organizational and operational processes 4. Implement operational models using a process aware information system 5. **Create KPIs (Key Performance Indicators) based on the previously modeled goals** 6. Once defined the KPIs, define the measures (particular way of measuring an indicator) 7. Implement measurement using IT systems
Evaluation Phase (Bottom-up Methodology)	1. Compute the measures on the basis of execution data 2. **After measuring, calculate the KPIs values and aggregate these values** 3. **Evaluate the level of goal satisfaction (and indirectly the SLA of business service at the business level)**

In the sequel, in order to enhance the current BPM methodology, we first describe the current state of art in each phase with its respective shortcomings. Afterwards, we introduce the additional methodological steps and relate them with the existent ones. Since we argue that processes must be increased with the modeling of business services and goals in the **Business Process Design and Analysis Phase**, besides describing the *methodological steps* related with these activities in the BPM design phase, we also give a brief overview of how these concepts can be modeled by describing the *languages for their representation* in other topics of research.

A. Business Process Design and Analysis Phase

The BPM lifecycle starts with the **Process Design Phase**, in which business processes are identified and represented accordingly. Business process models are recognized to be central in many approaches since they bring together all the *architectural elements*, organize them into a sequence of tasks or activities ("the process") in order to deliver business value [8]. Different levels of abstraction can be identified for business processes, ranging from high-level organizational processes to implemented processes [1]. The organized overview of all the business processes composes the so-called *process architecture* [13] or *process landscape* [1]. Once one has the process architecture in place, the business processes themselves can then be modeled using a particular notation, as for example the BPMN language [14].

The design of the process is guided by the following questions: "who does what, in what sequence, what services and products are produced, and what software systems and data are used to support the process?" [15]. The processes are then represented, with a focus on the activities that are required for achieving the desired results. In the cases where processes have already been modeled, changes in the process design can be made based on previously identified improvement possibilities.

Afterwards, the processes must be validated to check their correctness properties and simulated to detect possible undesired execution sequences [1]. Processes must be validated not only to address syntactical/structural properties, but they must also be checked with respect to their content, for example, whether all valid process instances are reflected in the corresponding process model or if processes only contain activities that generate value for the organization. In the cases that processes do not comply with this requirement, they must then re-designed (or re-engineered). In this context, *business process reengineering* approaches like “best practices” [16] and other methods like "classic" reengineering view [17] are suitable for addressing this concern.

The lack of an integration of process models with organizational services and goals in a SoEA motivated us to introduce activities related with the modeling of such concepts in this phase. We describe our approach as follows:

1) Additional Methodological Activities

The **Design Phase** of our methodology assumes a *top-down* fashion. We propose to integrate the concepts of service, goal and KPIs modeling together with process models. The first five activities of our methodology are performed in the design phase of the current BPM lifecycle, while the sixth step is performed in the configuration phase. In the sequel, we describe the required additional steps, its respective activities and how the current literature related to the field can be used.

Modeling Services and Goals. Goal modeling is the artifact employed for capturing the motivational aspect and strategies behind the organizational practices. Moreover, by adopting goal modeling, the company can systematically express the choices behind multiple alternatives and explore new possible configurations for an organizational setting [9].

Goal-Oriented methods for goal modeling arose in the Requirements Engineering area (GORE), with frameworks like i*/Tropos [9] [18], KAOS [19] and NFR framework [20]. In this area, goals are considered as *requirements* for a target system to be constructed. Enterprise architecture methods incorporated the same idea for modeling goals and other motivational aspects [10]. In this field, goals are considered stakeholders’ *requirements* for an architecture yet-to-be constructed or redesigned [21]. Some works in the BPM discipline have also been inspired by the GORE orientation [4] [22] [23] [24] [5] and proposed some extension of the traditional process techniques with the addition of goal-related concepts. This approach aims at overcoming the semantic gap between high-level enterprise’s goals and the business processes that are responsible for implementing these goals at the operational level.

As a general conclusion of the area, we may argue that methods in GORE are well developed, although they are suitable for representation of goals in the context of RE activities. On the other hand, methods in EA and BPM are more appropriate for the representation of goal-related concepts in accordance with processes and other architectural models, despite the incipiency in the area for modeling these goal-related concepts [10].

The methodological support for modeling goals is commonly denominated as goal analysis. Goal analysis comprehends the general term for a *top-down* approach of goal elicitation that encompasses the process of exploring the documentation for the purpose of identifying, organizing and classifying goals [25]. Most of research initiatives related to goals focuses on goal modeling and analysis in the context of GORE approaches, while the area of goal elicitation and discovery has remained largely neglected. As a result, goal elicitation remains a challenging activity with problems involving methodological guidance [26]. Among the methods that can used for goal elicitation and modeling, we may cite: the Goal Question Metric (GQM) [27], the Goal-Based Requirements Analysis Method GBRAM [28], NFR (Non-Functional Requirement) framework [20].

Regarding the literature for service description languages, although the service development discipline is still in its early stages [29], some trends can be identified. In the first trend, service description methods allow developers to describe how a web-based software-to-software service can be invoked [30] [31], by using the Web Service Description Language (WSDL). Based on the needs of the alignment of business/IT, some approaches incorporate the representation of more business-oriented aspects such as normative requirements (obligations, penalties, pricing, ownership, legal aspects among others), provisioning, composition and bundling in addition to the description of technical aspects of services [32] [30] [33]. In a third vein, we may find approaches on the fields of enterprise engineering and value modelling that recognize the nature of enterprises as services systems that aim at providing services to the external world. Among these, we can include the approaches found in [7] [29] [31]. This is precisely the view that we pursue in our approach, by modeling business aspects in which stakeholders consume organizational services and want to have their respective goals fulfilled.

We start our approach with the elaboration of the organizational goals on the basis of the services that are provided to the external world. These concepts can be better illustrated by means of a running example. For example, in the case of an insurance company, several types of insurance services are usually provided to its customers. In this case, the customer can be considered as the *service requester* for a *Life insurance service*, provided by the insurance company (*service provider*). For each organizational service, usually several goals are created. In this case, the goals in our approach represent the goals of the customer (*service requester*) and the goals of the insurance company (*service provider*). The customer delegates (or depend) of the *provider* to achieve some goals, and in exchange, the customer also has to fulfill some of the *provider’s* goals (for example, the *provider* needs a customer’s information and if the customer fails in achieving the provider’s goals, then the *service provider* either loses its ability of delivering the organizational services to the customer or it delivers with lower quality)[1]. Observe that in this case, the service may be associated to many goals that express either *functional* or *non-functional* requirements to the service. Furthermore, the normative aspect (norms and contracts) usually governs the relation among *requester* and *provider*, establishing a Service Level Agreement (SLA) between them.

Following with the description of the organizational goals (*provider's* goals), basically speaking, the organization exists in order to satisfy some of the stakeholders' goals. In our approach, the *provider's* goals have been elaborated on the basis of the services that justify the creation of the organization. These goals are by nature high-level goals (in many approaches they are denominated as *mission* ad *vision* [34]) and must be refined into lower-level goals by senior management until reach the operational level, i.e., the *architectural elements* (business processes, resources, roles, data, etc.) that are responsible for achieving them.

Once the organization is an already operating environment, these goals are usually implicit into the organizational scenario, but they must be modeled accordingly. Usually, goal formulation is an activity of high complexity in real-world scenarios and stakeholders may elaborate them in several levels of abstraction and precision [35]. Among the sources which could potentially provide goals for analysts, we may cite: (i) stakeholders who can explicitly state them through interviews; (ii) preliminary material about the organization (iii) preliminary analysis of the current system current organizational setting) with the identification of problems and deficiencies which lead to TO-BE goals [36] [37]; and (iv) policies, strategies, products, processes, models of the organization [27] and mission statements [38] [39].

Once a preliminary set of goals has been identified (using the aforementioned sources), refinement and abstraction techniques can be applied [36], by breaking down a higher-level goal into several lower-level goals in such a way that the latter contributes to the achievement or satisfaction of the former. This is the process of goal refinement that comprehends the subdivision of a set of identified goals into a logical sub-grouping so that the requirements can be more easily understood, defined and specified [25]. Goals can be refined based on different criteria and using different techniques. For example, the refinement can be made based on the subject matter of the goal like in the NFR (Non-Functional Requirement) framework [20], by stating the reasons for the existence of the goal and using refinement/abstraction techniques (why/how questions) [36], by using AND/OR decomposition [9] [19], can be made based on the process architecture of the organization [13] and/or organizational structure [6]. During the process of goal refinement, naturally conflicts among goals may arise [39] [40] and must be solved, possibly using priorities among goals.

Furthermore, goals also need to be characterized with respect to how their satisfaction is checked for a given time interval. For example, some goals in a given time interval must be achieved, while other must be prevented. This time interval for their satisfaction need also to be set up as well as their owners. Following with our example, after an organizational evaluation, the insurance company knows that its refund benefit process is not currently performing in a very efficient manner and for this reason, it is desired to increase this performance (*Increase the efficiency of process of refund benefits* goal). On the other hand, the organization knows that its admission process for new customers is performing well and for this reason, the *Maintain the level of new customer admission* goal is also elaborated.

After the goal refinement process, the goals reach a level of abstraction that can be operationalized. Goal operationalization is the process of relating goals to operations that ensure their fulfillment [25]. In the context of architectural models, the "operations" are represented by the *architectural elements* [41]. Among these architectural elements, business process receives a special attention since they represent the link between the goals and the remaining *architectural elements*. This operationalization can be done in two ways: (i) by relating goals to *architecture artifacts* and/or (ii) by relating goals and *architecture artifacts* via intermediate goals [42]. Analogously to the process of goal refinement, conflicts in goal operationalization may arise and priorities may show to be helpful to manage such conflicts.

Ideally, all the *architectural elements* of an enterprise architecture should be traced to business goals ad vice versa, but in practice, this is not the case [40]. Therefore, the *architectural elements* that do not have an associated goal should be eliminated. Conversely, for the goals whose any operational *architectural element* can be linked to, a feasible implementation for them must be found. In many cases, however, this operationalization may be not feasible implementable or even not easily implemented in a reasonable period of time. The organization must decide according to its policies and constraints which goals must be operationalized and in which order.

Modeling KPIs. Once the goals are operationalized accordingly, their satisfaction must be checked. Performance requirements on the enterprise architecture are specified through Key Performance Indicators (KPIs) with target values that must be reached in a certain time period [43]. The purpose of defining KPIs is to measure in each extent the (operationalized) goals are currently being achieved.

Concerning the KPIs modeling, the BPM area presents several approaches that address the topic KPI representation (through *modeling languages* for KPIs) [44] [45] [46] [47] [48]. A very recent effort for KPI modeling is proposed in [44] that builds its contribution on the basis of extensive literature review. As can be deduced from this work, the literature in KPI modeling depicts a number of problems regarding (i) the visual (graphical) representation of the KPI on the process models in which they are defined and (ii) the expressiveness regarding the definition of common types of

[1] This idea of *goal dependency* has been inherited from the Tropos framework and modeling language [18].

KPIs usually found in real-world scenarios. The proposal addresses these issues, but the topic of visual representation of KPIs on the basis of business processes and other architectural models still requires a lot of research investments [44] [49]. Enterprise architecture is also an area that presents some approaches for modeling indicators [39] [50], although very few approaches propose a modeling method for indicators as well as their integration into modern enterprise modeling frameworks [39] [50]. Therefore, this absence of appropriate KPI modeling language hampers efforts of integrating KPIs into the whole enterprise management lifecycle.

Regarding methodologies for collecting and modeling KPIs, we may basically divide the approaches into two groups, namely (i) approaches that are mainly originated in management sciences and enterprise management and (ii) methodologies from the BPM area.

The first group of proposals (management sciences and enterprise architecture) is concerned about the development of the so-called Performance Measurement Systems (PMS)[51]. There are basically two types of methodologies for developing PMS [15], that is: (i) to develop the indicator system based on existing generic indicators and/or complete PMS libraries and (ii) to develop the indicator system selecting the appropriate indicators for the company on the basis of its business objectives and success factors, resulting in a list that it is company-specific. In the first type, several PMS libraries for the development of PMS within a company-wide scope are available, such as the best-known Balanced Scorecard [52], Performance Measurement Matrix [53], the SMART Performance Pyramid [54] [55], the Performance Prism [56] and EFQM Business Excellence Model [57]. These libraries have a number of indicators and a methodological perspective that enables the library users to derive indicators for a specific organization, adapting the questions suggested in the library to some specific organizational context. The challenge consists in selecting the appropriate indicators from an extensive list of potential indicators. Furthermore, other disadvantage refers to the fact that the libraries provide indicators for determining the performance of the entire company or organization units (the indicators are very high-level and related with several business processes at the same time), in opposition to the evaluation of individual business processes [15] [58].

The second type of methodologies is based on the evidence that it seems very unlikely that a universal set of performance indicators can be applied successfully to all business processes. For this reason, some papers propose methodologies for derivation of process-specific indicators on the basis of goals. In contrast to the use of libraries, these methodologies: (i) develop PMS to evaluate the performance of business processes (in opposition to measure the performance of entire corporations or organizational units) and (ii) are related to the goals of each business process [15] [58]. A characteristic of these two types of approaches is that usually they operate in a *top-down* fashion, by operationalizing strategies and goals ("What do we want to measure?"). The computer support of this kind of performance management is usually achieved through the employment of Decision Support Systems (DSS), Executive Information Systems (EIS) or Business Intelligence (BI) projects that use Data Warehouse and OLAP tools [44], such those presented in [59] [60].

The proposals in the context of BPM initiatives are based on the fact that KPIs should be modeled together with business processes during the design phase. This practice allows the detection of dependencies amongst KPIs at design time and enables their usage as part of the business process analysis, for instance in business process simulation techniques [44]. The methodological aspect here is characterized by approaches that model process-specific indicators in a *bottom-up* fashion (by finding measurable aspects of processes, such as properties of activities and data objects or by using mandatory indicators that are provided by the management [50]). Other common characteristic of these approaches is that they abstract from the business content of the processes in question, opposing to PMS development approaches. It means that, a person in charge of eliciting and modeling the KPIs does not necessarily need to be a business subject matter expert [3].

The evaluation of indicators is done using technologies that are generally denominated as Complex-Event Processing technologies [61]. These approaches are commonly divided into three groups, depending on the moment that KPIs values are evaluated. Techniques from the *process controlling* [15] and *Process Mining* (PM) [62] fields allow one to perform *post analysis* of already finished process, whereas *Business Activity Monitoring* (BAM) approaches [45] [63] enables real-time monitoring of processes. Furthermore, *Process Intelligence* (PI) techniques [48] [64], data warehousing and classical data mining can be applied to perform predictive analysis on the business processes. They are based on the application of business intelligence techniques [48] (in particular data/process mining, data warehouse and OLAP tools) [44].

To summarize this discussion, we present the three main groups for approaches in eliciting, refining and modeling KPIs in literature (depicted in Table II).

Table II Characteristics of KPI methodologies

<table>
<tr><th>Approaches</th><th>Characteristics of the Approach</th></tr>
<tr><td rowspan="2">Management Sciences [52] [53] [54] [55] [56] [57] and Enterprise Management [27] [65] [50] [39] [15] [58]</td><td>• Organizational-wide
• Use indicators libraries
• Top-down fashion
• Measurement using BI and Datawarehouse tools
• Using informal descriptions, with no graphical representation of indicators. Exceptions can be found at [39] [50]</td></tr>
<tr><td>• Process-specific
• Goal-based (althouth goals are not explicitly represented in models)
• Top-down fashion
• Measurement using BI and Datawarehouse tools
• Using informal descriptions, with no graphical representation of indicators</td></tr>
</table>

Approaches	Characteristics of the Approach
BPM [44] [45] [46] [47] [48]	• Process-specific • Bottom-up fashion • Measurement using Complex-event Processing tools • Modeling methods strongly implementation-oriented (few business-oriented solutions [44]) and proprietary solutions [48]

A brief evaluation of the methodologies and modeling languages for the KPI representation shows the need of integrating the discipline across several areas. Using the results found in literature, we propose the following requirements for an unified methodology that strive to model KPIs:

- The methodology must be goal-oriented (with goals explicitly represented using graphical languages);
- It must represent process-specific indicators, but also organizational-wide indicators. While the process-oriented indicators are more closely related with the obtainment of operational data to evaluate the organizational performance, the organizational-wide indicators may be helpful to give strategic insights for the management;
- Since it uses organizational-wide and process-specific indicators, the methodology assumes at the same time, a *bottom-up* and *top-down* fashion. In fact, both directions are not necessarily disjunctive and can be used together [50];
- The methodology may use the indicators libraries as inspiration for the organization to adapt general indicators that might be suitable to its needs and scenarios. Furthermore, by using reference models (libraries), one is able to adapt the indicator with respect to the semantics in the particular domain. This may be considered a good practice, since one of the crucial aspects in the process of building a PMS lies in the identification of the appropriate performance indicators [58].
- Considering the computer support for the measurement of the indicators values, the methodology may be benefited by the use of BI and Data warehouse tools, but also by Complex-event Processing technologies. Moreover, computer support for the KPI management can be done using dashboards or other performance management software that provide navigation/retrieval functionalities [50].

By incorporating the requirements of this methodology, we may propose a rough method to create KPIs to measure our goals. Before following with our running example, however, it is paramount to briefly delineate the conceptual difference between goals and KPIs. As in the previous section, goals are considered to be *requirements* in our approach, while a KPI is intended to measure the success (or failure) of the goal [39] providing critical information to the organization for monitoring and predicting business performance in accordance with the strategic goals [61]. This difference implies that, for example, if we chose the *Increase the efficiency of process of refund benefits* goal (requirement for the refund benefit process) from our running example, we must be able to numerically determine the current performance of this process to determine the level of achievement of this goal. In order to do that, two KPIs can be identified: *Number of evaluations of refund benefit* (characterizing performance in terms of *quantity*) or *Average time spent in the process* (characterizing performance in terms of *time*). One of the indicators (or both) may be selected according to one's needs depending on the conceptualization of performance (if one decides to conceptualize performance either as *time* or as *quantity*).

In this example, indicator libraries have not been used, but KPI discovery could be facilitated by their use. Furthermore, we created an indicator that is directly related and measured in one specific business process. This is not necessarily true in all cases, and in many situations it is necessary to create indicators (called as organizational indicators) that are measured either in several business processes or by using BI techniques.

B. *Process Configuration Phase*

Once the operational processes have been modeled and verified accordingly, the **Process Configuration Phase** investigates how these processes can be implemented with the support of technology. This realization involves the selection of a suitable platform to run business processes, and subsequent implementation of the processes through the configuration of a process aware information system (e.g. workflow management system) [12], in a Service-Oriented Architecture (SOA) [1].

In a SOA, the technical implementation of business processes is usually accomplished by translating activities from the process level to the workflow level, using the Business Process Execution Language (BPEL), web services or other SOA platforms (other appropriate workflow engines) [46] [1]. Further, the organizational environment also needs to be represented in the case of human interaction with the workflow system as well as integration with external applications, standardized ERP systems, SCM systems or CRM systems [15] [1].

Besides the technical implementation of business processes, the KPIs also need to be technically implemented. This involves two actions [44]: first, it is necessary to decide which measures must be taken during the execution of the process to calculate the KPI (instrumentation). This instrumentation represents the specification of how the KPIs must be measured in the business process models through the specification of *monitoring points* [1] [46] [66]. Monitoring points define points in the process at which data comes from for the KPIs computation [66]. After configuring the monitoring points, specific deployment information on which events have to be logged at runtime must be also specified [45] by configuring the event log of the information system appropriately or by using an API of the information system to query its history [44]. Alternatively, the events can also be published on a publish/subscribe infrastructure to enable real-time monitoring [45].

Currently, some approaches allow one to define these measurement points to gain traceability, although in many cases, there are no evidences that this traceability can be maintained in both directions [44]. By modeling goals, services and KPIs, our methodology helps to connect the business services with the implementation of the organizational services using processes and IT systems.

Following with the running example, we decided to choose the *Number of evaluations of refund benefit* KPI and implement it in the refund benefit process. The KPI has been defined in the previous phase and the next step is to decide how to calculate it from the data that is generated during the execution of the process (by implementing the monitoring points). In the case of *Number of evaluations of refund benefit* KPI; the *number of evaluation* measure may correspond to the different measures within the refund process. For example, when the process of refund executes, it issues whether a given request of refund is accepted or not. Therefore, the *number of evaluation* measure may correspond to all requests submitted in a time interval of 1 month. Alternatively, it may correspond to only those requests that have been made by the customers with less than 60 years old in a time interval of 2 months. As during the creation of KPIs, the measures that are created on the basis of the KPI depend on the business needs. For instance, when data about running processes are not automatically obtained or the costs of measuring are too high, only a representative portion of monitoring points of interest may be selected (and thus to obtain one particular measure is cheaper than the other).

Since we decided to implement the *number of evaluation* measure that correspond to all requests submitted in a time interval of 1 month, we have to set other two parameters: (i) the target value: while our goal states that the performance should be increased, it does not specify a quantitative value for that. So, we decided based on business needs that, in order to increase the performance, the current value (that currently is number of evaluations = 80 (evaluations)) must be increased and we set up the target value for *number of evaluation* measure > 100 (evaluations); (ii) the correspondent monitoring points are then specified in the exact point of measurement in the process. In this case, we attach one monitoring point in the start of the process in order to know how many times the process has been started (how many instances have been created). The second action is to decide how to obtain the measures that are necessary to calculate the KPIs. Since the key monitoring points track information about business processes through monitoring the occurrence events, a BPEL engine that support the refund process may be configured in this case.

C. Business Process Execution Phase

The next phase is the **Business Process Execution Phase**, which encompasses the execution of the implemented processes. In a SOA, implemented processes are usually realized through orchestrations using service-composition techniques [1]. The interaction between process participants generate events that are stored in the logs captured by the underlying IT infrastructure such as ERP, CRM and Workflow Management systems [67] configured in the previous phase.

Based on these events, KPIs' values are calculated and displayed in dashboards [44] [45] [46]. Observe that service provider and requester agree on quality of service in advance and the KPI values are documented through a SLA. In a SOA architecture, the main approaches for calculating the KPIs are event-based like those presented in [44] [45] [46] [66] [67] [68] [69], but it is possible to find other ways of performing such measurement, for example, using formal techniques for analysis of executions of organizational scenarios [39] or using data mining techniques [48], BI/ Data Warehouse technologies [50] [70] [71].

After an evaluation of this phase, we may conclude that KPI measurement is currently characterized as being highly *operational* and mainly focused at the implementation level, rather than in the business level. This can be accounted by the fact that the approaches in the context of event-based technologies mainly rely on the computation of (low-level) measurements about web-services composition execution [72] [73]. Exceptions can be found at [70] [71] that link the evaluation of goal satisfaction to indicators, although goals are not faced as business services to be provided to the external environment. We discuss in more detail in next section how to use the approaches for calculating KPIs and how to integrate them with our overall methodology.

D. Business Process Evaluation Phase

Finally, in the **Business Process Evaluation Phase**, events generated by process execution can be analyzed in order to reveal areas for potential improvement, enabling the obtainment of insights about qualitative and quantitative aspects of business processes, also providing root-causes of undesired measures.

Basically, very few approaches meet the requirement of using execution data to calculate the process properties and subsequently use this data to make improvements in the process [2]. Among them, we have found the IBOM platform [43] that performs intelligent analysis on business measures to understand causes of undesired values and predict future values.

As a general conclusion, we may argue that research in this area is very much driven by practical requirements and tends to take an operational and short-term view. Exemplary work includes [48] [66] and reflects the close association of this field to industry and tool vendors [3]. Other approaches use goal-oriented analysis and how the satisfaction of goals can be evaluated on the basis of KPIs in [39] [70] [71]. We consider our approach to contrast with this general trend, once we take a long-term and business-oriented perspective by linking services and goals with the business process that implement such goals.

1) Additional Methodological Activities

This scarcity of methodologies that link operational aspects with the business concept of service led us to suggest additional steps into the BPM lifecycle. This phase of our methodology assume a *bottom-up* fashion since it collects operational data and evaluates the level of satisfaction of goals and business services based on these data. The phase started in the BPM execution phase and is responsible for collecting the operational data and calculating the SLAs on the basis of the IT events (web-services compositions). The current techniques for SLA measurement and calculation like presented in [72] [73] may be reused, and KPIs must be calculated and aggregated accordingly. With the KPIs values in hands, it is possible to evaluate the level of goal satisfaction. In the cases where goals are not satisfied in a reasonable level, the causes must be found. It is then possible to discover about the quality of services at the business level and whether the SLAs are being met.

To finalize our running example, we may decide to have a real-time or a *post-analysis*. Since the evaluation period of the indicator is 1 month, then we decided to evaluate it a *posteriori*. For the evaluation, the BPEL engine tracks all the evaluations that are submitted in a period of 1 month and we checked that, since the number of evaluations = 81 in this period, then our goal of increasing the performance is not achieved. If on the contrary, the number of evaluations > 100 in this period, then the goal would have been achieved and we could eliminate it from our model. In order to evaluate the level of achievement of other goals (not discussed here), we could use mechanisms for propagating goal satisfaction values through the goal model. Finally, the quality of service between the insurance company and the customers can be evaluated in terms of the achievement of their business goals.

A last remark in this phase refers to possibility of eliminating the goals and KPIs of their respective models. As mentioned before, goals may disappear because either they have been successfully achieved or because they have been identified as no longer essential. The previously specified KPIs may also change because either an evolution of the business process or their own evolution (as a result of goal creation).

III. Discussion

This paper enhances the current BPM lifecycle with the explicit modeling of business services and goals as means of capturing a more strategic view for processes. We have been working in this topic of alignment of business goals with processes and other architectural models in previous efforts such as in [74] [6] [75], striving to tackle this problem from the perspective of the representation (*language*) of goals in business processes and other enterprise models. After these efforts, the current paper is the realization that *methodological* support is also necessary to gain this more strategic perspective and a first effort into introducing these methodological aspects into the BPM lifecycle.

By the discussion provided about service and goal modeling, our approach is intrinsically *top*-down in the design phase, using the customer needs as input of our methodology and *bottom-up* in the measurement phase where it verifies whether these goals and services are adequately implemented. It is certainly true that some issues in the areas of goal and service modeling may hamper the success of our methodology, as for example the translation of the high-level strategic goals until reaching the operational level that is considered to be a difficult task. Nevertheless, we believe that this approach can be valuable not only to connect the customer perspective with the internal operations of the organization (as defended in this paper), but also to increase the value of other already existent activities in the context of process management as for example, to guide the development process of the *process architecture* or the activities performed in the context of *business process reengineering* approaches as described in the BPM design phase.

As a general conclusion, we may argue that this approach is valuable from both the BPM and EA perspectives. From the BPM perspective, there is a huge gap between linking business strategies to the management of operational performance from a methodological point of view (also noticed by [3]). Our approach thus is valuable to enable process improvement efforts to be more connected with the customer perspective, directly addressing the customer needs. From the EA perspective, this also aggregates value as it connects the enterprise viewpoints and resources to the business processes that operationalize them. Our claim is not to replace the methods and languages in any of the disciplines, but rather join them to obtain the aforementioned advantages.

As limitations for our approach, we may cite the absence of languages that fully support the concepts that we propose to capture here. To have a language that capture goals and services in the context of process models is certainly a future work that should be addressed to enable such approach. Moreover, the methodologies that we mentioned in each phase of our approach should also be developed towards achieving this overarching goal, once they are still incipient. These are certainly two veins of research that we intend to address as our future steps. Finally, the application of the proposed methodological steps in real-world scenarios is fundamental to validate our contribution in practice. In that respect, we have pursued this goal in a previous approach [6], but the development of more cases studies is certainly a valuable path to enforce the ideas suggested along this paper.

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